

## GOVERN (GV) — Organizational Oversight & Accountability

The AI Risk Management Framework (RMF) aims to establish governance, roles, accountability, and oversight.

GSP Alignment:

- G-BALO Role-Based Governance: This replaces the non-personal, replaceable governance model.
- Independent Certification Bodies: These bodies provide independent certification.
- Conflict-of-Interest Controls: These controls ensure transparency and prevent conflicts of interest.
- Appeals and Revocation Mechanisms: These mechanisms allow for appeals and revocation of certifications.
- Founder Independence & Non-Reliance Clause: This clause ensures that the founder's independence and non-reliance are maintained.

Key Signal to NIST:

Governance is structural and auditable, not policy-based or aspirational.

## MAP (MP) — Context, Use, and Risk Framing

The AI RMF aims to understand the system context, stakeholders, and risks.

GSP Alignment:

- Attestation Metadata Captures: These capture system context, operational environment, user/system/developer action boundaries, and sector-specific use-case mappings (healthcare, defense, finance, infrastructure).
- Key Signal: Risk context is captured at runtime, not inferred post hoc.

## MEASURE (ME) — Risk Measurement & Evidence

The AI RMF aims to quantify and document risks and impacts.

GSP Alignment:

- Cryptographic Moral Receipts: These provide verifiable evidence of risk measurement.
- Hash-Chained Attestations: These ensure the integrity and authenticity of attestations.
- Reconstruction Capability: This allows for the reconstruction of the system's state.
- Structural Integrity Index Metrics (SIS, RCS, CCIS): These metrics assess the system's structural integrity.
- Key Signal: Measurement is verifiable evidence, not self-reporting.

## MANAGE (MG) — Risk Response & Mitigation

The AI RMF aims to respond to risks, incidents, and failures.

GSP Alignment:

- Immutable Audit Trails for Incident Response: These provide a clear and auditable record of incident responses.
- Fault Attribution (User vs System vs Developer): This helps in identifying the source of faults.
- Post-Certification Spot Checks: These ensure that certifications are maintained and up-to-date.
- Revocation and Remediation Workflows: These workflows facilitate the revocation and remediation of certifications.
- Key Signal: Risk management is operationally enforceable, not advisory.

NIST Summary Sentence (Use This)

“The Glass Substrate Protocol operationalizes the NIST AI Risk Management Framework by embedding governance, measurement, and accountability directly into system execution and evidence infrastructure.”

This sentence should be included in your submission packet.

## 2. IEEE Working Groups — Exact Targets (No Guessing)

IEEE does not recognize “AI governance” as a category. You must identify the appropriate working groups.

Primary IEEE Targets:

- IEEE P7000: This standard explicitly addresses processes for ethical systems. The Glass Substrate Protocol provides the technical enforcement layer that these standards currently lack.
- Pitch angle: GSP supplies hardware-enforced auditability for P7000 governance processes.
- IEEE P7009: Bias standards fail without reconstruction capability.
- Pitch angle: GSP enables post-hoc bias investigation with cryptographic evidence, not statistical inference alone.
- IEEE P7010: Well-being claims require accountability.
- Pitch angle: GSP provides traceability between AI actions and human impact claims.

Secondary / Adjacent IEEE Targets:

- IEEE Systems Council (systems assurance)
- IEEE Reliability Society (fault attribution, failure reconstruction)
- IEEE SA Industry Connections (pre-standard incubation)

IEEE Positioning Sentence:

“GSP is not an ethics framework; it is an enforcement substrate that enables existing IEEE ethical and safety standards to be auditable, reconstructable, and legally defensible.”  
This sentence is crucial.

### 3. Call for Independent Certification Bodies (Publish This)

This document serves as the catalyst for the ecosystem.

It should reside in the /certification/ directory and be linkable in submissions.

Title:

Call for Independent Certification Bodies — Glass Substrate Protocol (GSP)

Purpose:

To establish a global, independent network of certification bodies authorized to assess conformance with the Glass Substrate Protocol.

Eligibility Criteria (Objective, Not Discretionary):

Applicants must demonstrate:

- Legal independence from AI system developers
- Expertise in at least two of the following:
  - Systems security
  - Cryptography
  - Safety certification
  - Regulatory compliance
- Published conflict-of-interest policy
- Ability to perform technical audits and evidence review
- No prior relationship with G-BALO is required.

Application Materials (Self-Service):

Applicants submit:

- Organizational charter
- Audit methodology
- Conflict-of-interest disclosures
- Appeals handling procedure
- Designated technical leads

Evaluation Process:

- Completeness review (objective)
- Technical capability assessment
- Provisional authorization (time-limited)
- Mandatory third-party audit within the first year
- No founder approval. No discretionary veto.

Ongoing Obligations:

- Annual independent audit

- Participation in interoperability testing
- Adherence to public transparency requirements
- Acceptance of revocation for non-compliance

*Authorization as a GSP certification body is non-exclusive. Multiple independent bodies may operate concurrently across jurisdictions and sectors.*

*The Glass Substrate Protocol is designed such that trust emerges from evidence, independence, and repeatability, not from centralized authority.*